

Resource Scheduling Framework with Priority Queue Optimization

INTELLIGENT RESOURCE MANAGEMENT
Resource Scheduler Pro

SYSTEM READY

Plan work. Schedule smarter.

Run an intelligent priority-queue scheduler that balances urgency, deadlines, waiting time and resource requirements.

Priority QueueDeadline AwareStarvation ControlResource Insight

▶

Start a scheduling session

Enter basic session details to begin.

Operator name

Project / Department

Scheduling strategy

Enter Scheduler →

Session data is stored locally for this browser.

LIVE SCHEDULING WORKSPACE

Schedule smarter.

The original project uses priority-based scheduling. This enhanced implementation adds smart scoring, starvation prevention, deadline awareness, resource visibility, simulation metrics and an explainable decision layer.

New Session▶ Run Smart Scheduler

OPERATOR
John

PROJECT
Cloud Operations

STRATEGY
Smart Priority Scheduler

STATUS
Ready

TASKS
3
total workload

AVG WAIT
10 min
average delay

AVG TURNAROUND
20 min
completion time

THROUGHPUT
6 tasks/hr
processing rate

UTILIZATION
75%
resource usage

Add Task

Priority Queue

Task name

Priority (1-10)

Execution (min)

CPU %

Memory MB

Live Queue

Clear

1student report
General • P9 • 10 min • CPU 20% • 512 MB
SMART SCORE
110.1

2database backup
General • P5 • 10 min • CPU 20% • 512 MB
SMART SCORE
70.1

3CPU backup
General • P5 • 10 min • CPU 20% • 512 MB
SMART SCORE
70.0

localhost:8080

Paused

Task name

CPU backup

Priority (1-10)

5

Execution (min)

10

CPU %

20

Memory MB

512

Category

General

Deadline

dd-mm-yyyy --:-- --

+ Add Task

1 student report

General • P9 • 10 min • CPU 20% • 512 MB

SMART SCORE

110.1

2 database backup

General • P5 • 10 min • CPU 20% • 512 MB

SMART SCORE

70.1

3 CPU backup

General • P5 • 10 min • CPU 20% • 512 MB

SMART SCORE

70.0

Execution Timeline

Waiting for schedule

Run the scheduler to generate an execution plan.

Why this task ran first

Explainable scoring

Add tasks and run the scheduler. The system will show exactly how the selected task earned its score and why it was chosen over the remaining tasks.

Execution Timeline

Scheduled 3 tasks successfully

Slot 1 • 0-10 min

student report

Score 110.1 • SCHEDULED

Slot 2 • 10-20 min

database backup

Score 70.3 • SCHEDULED

Slot 3 • 20-30 min

CPU backup

Score 70.0 • SCHEDULED

Why this task ran first

Explainable scoring

✓ student report was selected first because it had the highest smart score among all tasks waiting in the queue.

Selected at position 1 because it had the highest smart score among the remaining tasks. Base priority contributes 90.0; deadline urgency +0.13; waiting/aging contributes +20.0. Final smart score = 110.13.

Priority 90.0 Deadline urgency +0.1 Aging bonus +20.0 Final score 110.1 Execution 10 min

1 student report

Score 110.1 • Wait 0 min

Selected at position 1 because it had the highest smart score among the remaining tasks. Base priority contributes 90.0; deadline urgency +0.13; waiting/aging contributes +20.0. Final smart score = 110.13.

2 database backup

Score 70.3 • Wait 10 min

Selected at position 2 because it had the highest smart score among the remaining tasks, while preserving its aging contribution of +20.0. Base priority contributes 50.0; deadline urgency +0.25; waiting/aging contributes +20.0. Final smart score = 70.25.

3 CPU backup

Score 70.0 • Wait 20 min

Selected at position 3 because it had the highest smart score among the remaining tasks, while preserving its aging contribution of +20.0. Base priority contributes 50.0; No deadline bonus; waiting/aging contributes +20.0. Final smart score = 70.0.

View Final Results →

✓ SESSION COMPLETE

Scheduling completed.

Cloud Operations was processed for John using Smart Priority Scheduler.

← Dashboard

Start New Session

TASKS PROCESSED

3

AVG WAITING

10 min

AVG TURNAROUND

20 min

THROUGHPUT

6 tasks/hr

UTILIZATION

75%

STARVATION CONTROL

ACTIVE

Task execution summary

Completed Plan

TASK	SCORE	WAIT	TURNAROUND	STATUS
1. student report General	110.1	0 min	10 min	✓ SCHEDULED
2. database backup General	70.3	10 min	20 min	✓ SCHEDULED
3. CPU backup General	70.0	20 min	30 min	✓ SCHEDULED

Scheduler insight

SMART



Smart ordering was applied

student report led the queue because its combined score considered base priority, deadline urgency and aging. The decision trail records the exact reason for every selection.

✓ Deadline awareness

✓ Aging / starvation prevention

✓ Explainable selection

Algorithm comparison

Performance view

✓ Explainable selection

Algorithm comparison

A baseline comparison using the same workload demonstrates the benefit of priority-aware ordering.

FCFS Baseline

10.0 min

Average waiting time

6.0 tasks/hr

Throughput

Priority Queue

10.0 min

Average waiting time

6.0 tasks/hr

Throughput

Smart Scheduler

10.0 min

Average waiting time

6.0 tasks/hr

Throughput

Generate Session Report

Run Another Session →